The U.S. Multinational Advantage during the 2008-2009 Financial Crisis: The Role of Services Trade

by

Fariha Kamal
Federal Reserve Board

Zachary Kroff
Analysis Group

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Abstract

We document the augmenting role of services exports in U.S. multinationals' goods-export growth during the global financial crisis. Using newly linked data on U.S. firms' foreign sales of goods and services and a triple-difference identification strategy combined with propensity-score matching, we find that compared to multinationals that only export goods (mono-exporters), multinationals that also export services to the same destination (bi-exporters) experienced higher goods-export growth. This result is driven by sales of intellectual property rights related to industrial processes (e.g., patents, trademarks). We also find higher growth in bi-exporters' foreign affiliate services sales and domestic employment in services sectors. These results reveal a pivotal role of services exports in supporting foreign demand for U.S. goods during the crisis.

Keyword: multinationals, financial crisis, services exports, goods exports

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* Contact: fariha.kamal@frb.gov; zachary.kroff@gmail.com. Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau, U.S. Bureau of Economic Analysis, U.S. Department of Commerce, Board of Governors of the Federal Reserve System, or their research staff. Kroff is an advanced analytics specialist at Analysis Group. The views presented in this article are those of the authors and do not represent any opinions or positions of Analysis Group or its clients. The data analysis was completed while Kamal and Kroff were economists at the U.S. Census Bureau. The Census Bureau and Bureau of Economic Analysis have ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 6907751: CBDRB-FY20-CES007-007, CBDRB-FY21-CES007-009, CBDRB-FY22-CES007-011, CBDRB-FY23-CES007-03, BEA-FY21-P6907751-R1, BEA-FY22-P6907751-R5, BEA-FY23-P6907751-R1). We thank James Fetzer, Alexis Grimm, Raymond Mataloni, Jessica McCloskey, and Christopher Steiner for guidance with BEA surveys; Laura Alfaro, Vanessa Alviarez, Andrea Ariu, Emek Basker, Kerem Cosar, James Harrigan, J. Bradford Jensen, Shawn Klimek, Isabela Manelici, Raymond Mataloni, John McLaren, Christian Volpe Martincus, Sean Wang, Nikolas Zolas and seminar participants at the Census Bureau, AEA, APTS, CAED, ELSNIT, ETOS, FIW, Kent State University, SEA, USITC, University of Alberta, UVA, IADB for valuable comments and discussions.

1 Introduction

The 2008–2009 financial crisis was marked by a precipitous decline in global goods trade (Baldwin, 2009). Notably, between 2007 and 2009, U.S. merchandise exports declined by 1% for multinational enterprises (MNEs) and 32% for non-MNEs (authors' calculations from U.S. Census Bureau, 2023; U.S. Bureau of Economic Analysis, 2025a). This paper examines multinationals' goods-export growth during the global financial crisis to understand the multinational advantage.

We propose a novel channel: complementarity between goods and services exports. The provision of services may enable firms to boost goods sales, in particular, driven by one-way demand complementarity where the good is essential but the service is optional.[1] Goods that may feature one-way complementarity with services include cellular communication devices, computers, routers, and switches where a service like computer software (e.g. Cisco's AnyConnect Secure Mobility Client, Microsoft 365) cannot be used without the good;[2] or General Electric Company's Predix platform that provides digital and data analytics services to monitor the performance of the company's jet engines or wind turbines to "creat[e] efficiencies for its customers that go beyond the physical engine." (General Electric, n.d.).

We test whether multinationals' sales of services to the same destination as their goods exports contributed to the goods-export growth advantage during the global financial crisis motivated by two facts. First, multinationals mediate over 90% of U.S. services trade (Bruner and Grimm, 2019). Second, U.S. services trade was especially resilient during the global financial crisis. Monthly U.S. goods trade declined by about one-third between July 2008 and February 2009 and the decline in services trade over the same time period was only about one-tenth (Borchert and Mattoo, 2009). Within U.S. services exports, transport and tourism declined; however, insurance, financial, telecommunications, business, and professional and technical services all grew.

Given multinationals' outsized participation in U.S. services trade, we focus on multinational goods exporters and construct propensity scores of the likelihood that they also exports services. We match multinationals that only export goods to a given destination (mono-exporter) with observable characteristics similar to multinationals that sell both goods and services to the same destination (bi-exporter) in the pre-crisis period. Employing a

[1]The idea follows Ariu, Mayneris and Parenti (2020) where consumer utility only increases when the service is associated with a good but the service itself does not raise consumer utility.

[2]Cisco AnyConnect Secure Mobility Client is a security endpoint software product that allows access to support remote users across wired and wireless connectivity and Virtual Private Network (VPN) connection (Cisco Systems Inc., 2023a); Microsoft 365 is a product family of productivity software, collaboration and cloud-based services owned by Microsoft Corporation (Microsoft Corporation, 2023).

triple-difference (DDD) identification strategy, we compare bi- and mono-exporters' growth in goods exports, to countries with higher and lower GDP growth, before and after the financial crisis. The DDD estimate indicates that bi-exporters exposed to an average decline in foreign demand in the top 15 export destinations experienced a 6.6 percentage points (pp) higher goods-export growth than matched mono-exporters. This result is robust to an alternate control group and varies systematically by destination-specific services trade barriers.

We find evidence of one-way complementarity augmenting bi-exporters' goods-export growth. We begin by showing that bi-exporters' higher goods-export growth is driven almost entirely by exports of intellectual property (IP) rights related to industrial processes such as patents and trade secrets. Guided by this result, we classify high-intangible goods that are most likely to exhibit one-way complementarity with industrial IP such as smartphones or data processing machines. We find higher growth in high-intangible goods exports by bi-exporters compared to mono-exporters as the main driver of the baseline result.

Finally, we explore two implications of our main finding for bi-exporters' services activities during the crisis. First, we examine whether MNEs also provide services through their affiliates located in the goods buyers' markets. We find that services sales growth is higher at foreign affiliates located in countries where their U.S. parents export both goods and services than at foreign affiliates located in countries where their U.S. parents only export goods. This is consistent with bi-exporters selling services to the foreign buyer through their foreign affiliates. Second, we examine whether the crisis induced within-firm reorganization towards services activities by comparing domestic employment growth at bi- and mono-exporters. We find that bi-exporters experienced higher growth in services employment led by growth in the information and professional and technical services industries which map most closely to industrial IP services.

The analysis is made possible by newly constructed links between the U.S. Census Bureau's Business Register (BR) of the universe of U.S. employer establishments and the comprehensive multinational and services trade surveys collected by the U.S. Bureau of Economic Analysis (BEA). We identify U.S. parents and U.S. affiliates of foreign parents in 2007 using the firm-level surveys on U.S. direct investment abroad and foreign direct investment in the U.S., respectively. We define a firm's services export status in 2007 using sales of intellectual property (e.g., charges for the use of trademarks) and select business services (e.g, telecommunications, computer, information services, management consulting, research and development) that account for almost half of total U.S. services exports. We combine a firm's pre-crisis multinational and services export status with the universe of merchandise exports using confidential U.S. customs transactions from 2006 through 2009.

This paper makes three main contributions. First, we provide evidence of interconnections between goods and services exports for the world's largest services trader (World Trade Organization, 2024). Ariu, Breinlich, Corcos and Mion (2019) find that Belgian firms are more likely to source services and goods inputs from the same source country and that goods trade barriers dampen firm-level services imports from the same markets. Berlingieri, Marcolin and Ornelas (2021) find that French firms with more experience exporting goods to a destination are more likely to source export-related service inputs from these locations than from domestic sources. Ariu, Mayneris and Parenti (2020) show that services exports act as a demand shifter for Belgian firms increasing the perceived quality of their products. Our findings suggest that during periods of negative demand shocks, services may be an especially important channel to boost the demand for goods exports.

Second, we provide evidence consistent with servitization—the phenomenon where manufacturers increasingly sell services—led by U.S. multinationals.[3] Fort, Pierce and Schott (2018) document a trend in U.S. manufacturers' activities as "[...] an evolution of US manufacturing firms into 'neuro-facturers' that increasingly provide intellectual services rather than physical goods." Given multinationals' significant contributions to the U.S. economy—accounting for three quarters of manufacturing employment and more than 80% of all industrial R&D in 2017 (Foley, Hines, Mataloni and Wessel, 2021)—our results suggest that their adjustments to the global financial crisis contributed to the process of servitization in the U.S. economy.

Finally, we provide, to the best of our knowledge, the first firm-level evidence on U.S. multinationals and their goods-export adjustments during the global financial crisis. Alfaro and Chen (2012) compare the sales growth of foreign-owned and domestic-owned establishments across 53 countries during the financial crisis. They find that foreign-owned establishments performed significantly better during 2007–2008, but not in non-crisis years. Alviarez, Cravino and Levchenko (2017), using similar data across 34 countries, find that foreign affiliate sales growth was higher than domestic sales growth from 2004 to 2008 and lower for 2008–2009. Our focus, in contrast to these studies, is on goods-export growth of U.S. multinational parent firms in addition to the U.S. affiliates of foreign parents and the role of services exports in boosting the demand for goods exports.

[3]Evidence of servitization has also been documented in the United Kingdom (Breinlich, Soderbery and Wright, 2018), France (Crozet and Milet, 2017), and India (Grover and Mattoo, 2021).

2 Empirical Approach

We evaluate the role of services exports in shaping multinationals' goods-export growth during the global financial crisis. Because multinational and services exporting status are not random, we implement propensity score matching to construct economically meaningful counterfactuals based on observable characteristics. We match a MNE goods exporter that also exports services with an observationally equivalent MNE that only exports goods. Appendix A provides details on the propensity score model and balance tests. We then estimate a triple-difference specification as follows,

$$\begin{aligned}\Delta \ln Exports_{icq} = \beta_0 &+ \beta_1 \Delta \ln GDP_{cy} \times MNE^{SVC}_{ic,2007} \times I(> 2008Q3) \\ &+ \beta_2 \Delta \ln GDP_{cy} \times MNE^{SVC}_{ic,2007} + \beta_3 MNE^{SVC}_{ic,2007} \times I(> 2008Q3) \\ &+ \beta_4 MNE^{SVC}_{ic,2007} + \beta_5 \overline{\Delta \ln X_{ic,2005-2007}} + \alpha_{iq} + \alpha_{cq} + \alpha_{ir} + \varepsilon_{icq}, \end{aligned} \tag{1}$$

where $\Delta \ln Exports_{icq}$ is growth in firm i's arm's-length goods exports to country c in quarter q.[4] The base period is the same quarter in the previous year. For example, growth in firm i's goods exports to country c in 2007Q1 is $(\ln Exports_{ic5} - \ln Exports_{ic1})$ where 5 and 1 indicate 2007Q1 and 2006Q1, respectively. $MNE^{SVC}_{ic,2007}$ indicates whether firm i exported any services to country c in 2007. $\Delta \ln GDP_{cy}$ is country c's annual GDP growth in year y. $I(> 2008Q3)$ takes on a value of 1 after 2008Q3 and 0 otherwise informed by the timing of the collapse in goods trade (Baldwin, 2009). $\overline{\Delta \ln X_{ic,2005-2007}}$ measures firms' average growth in arm's length goods exports before the crisis and controls for pre-existing trends.

The included set of two-way fixed effects absorbs the individual coefficients on $\Delta \ln GDP_{cy}$, $I(> 2008Q3)$, and $\Delta \ln GDP_{cy} \times I(> 2008Q3)$. Firm-quarter, α_{iq}, fixed effects control for any time-varying unobservable firm-specific factors such as productivity; country-quarter, α_{cq}, fixed effects control for time-varying unobservable country-specific factors such as exchange rates or changes in foreign demand; firm-region, α_{ir}, fixed effects control for firm-specific unobservables that vary at the region level such as local firm presence or trading experience in that market.[5] β_0 is a constant. Robust standard errors are clustered at the country level.

The DDD coefficient, β_1, can be interpreted as the difference between two difference-in-difference estimators (Olden and Møen, 2022).

[4] We exclude MNE's sales to foreign affiliates since these transactions may be influenced by non-economic factors such as tax rates across jurisdictions.

[5] Regions are defined as North America, Central America, South America, Europe, Asia, Australia and Oceania, and Africa following U.S. Census Bureau (2021b).

3 Data

We identify firms' multinational status in 2007 using the direct investment surveys collected by the BEA and linked to the Census Bureau's BR (Kamal, McCloskey and Ouyang, 2022). The Annual Survey of U.S. Direct Investment Abroad (Form BE-11) identifies U.S. parents and their foreign affiliates and the Benchmark Survey of Foreign Direct Investment in the U.S. (Form BE-12) identifies U.S. affiliates of foreign parents (U.S. Bureau of Economic Analysis, 2025c,b). These surveys are the basis for official statistics on the activities of multinational enterprises (U.S. Bureau of Economic Analysis, 2018).

We identify firms' services export status in 2007 using the Quarterly Survey of Transactions in Selected Services and Intellectual Property with Foreign Persons (Form BE-125) linked to the Census Bureau's BR (Kamal and Ouyang, 2022). The BE-125 contains information on charges for the use of intellectual property; telecommunications, computer, and information services; and other business services such as professional and management consulting, research and development services, and technical services (U.S. Bureau of Economic Analysis, 2020a). In 2007, the services products collected in the BE-125 accounted for 43% of total U.S. services exports.[6]

We explore the role of different services products on goods-export growth by grouping them into five broad categories as follows:

- Industrial-related Intellectual Property (Industrial IP): includes rights related to industrial processes (patents, trade secrets, trademarks) computer software, franchise fees, and other intellectual property and intangibles.
- Media-related Intellectual Property (Media IP): includes rights related to film and television, books and sound recording, and broadcasting and recording.
- Business Services (Business): includes R&D testing, legal, accounting, management and consulting, advertising, architectural and engineering, construction, industrial engineering, mining, operational leasing, trade-related, sports and performing arts, educational and training, and other business services
- Telecommunication Services (Telecommunication): includes computer and data processing, telecommunication, and database and other information services.
- Other Services (Other): includes industrial-type maintenance and auxiliary insurance.

[6]The share of services surveyed in BE-125 in total U.S. services exports is shown in Appendix Table A-4.

For the average MNE in the sample, 35% of their total services exports is industrial IP; 1% media IP; 50% business services; 10% telecommunications; and almost 4% other services (Table 1).

We measure firms' goods exports using the Longitudinal Firm Trade Transactions Database (LFTTD) which links the near-universe (all exported shipments valued over US$2,500) of confidential merchandise trade transactions on customs declaration forms with the BR enabling identification of firms engaged in U.S. goods trade (Kamal and Ouyang, 2020; Bernard, Jensen and Schott, 2009).[7] We measure firm's annual growth in arm's length exports by destination and quarter between 2006 and 2009. The average MNE in the sample experienced a 1.5pp decline in goods-export growth (Table 1).

We construct annual GDP growth rate using GDP in constant 2010 U.S. dollars for 186 countries obtained primarily from the World Bank's World Development Indicators (World Bank, 2021b) and supplemented for select countries from the United Nations (United Nations, 2021) and the Federal Reserve Economic Data (Federal Reserve Economic Data, 2021). We exclude tax havens that may be used as a vehicle to lower multinationals' global tax obligations through income shifting across jurisdictions (Dyreng and Hanlon, 2021).[8] Shifting income through the movement of intangible assets such as intellectual property, which account for half of the services exports measured in the BE-125, is well documented (Jenniges, Mataloni, Stutzman and Xin, 2018). Thus, excluding tax havens increases confidence that the services export flows in the analysis capture real economic activity.

4 MNEs' Goods-Export Growth and Services Trade

Table 2, Panel A, reports the DDD coefficient from Equation 1. We find that bi-exporters exposed to a 1pp decline in GDP growth experienced a 1.2pp higher goods-export growth than matched mono-exporters in the post-crisis period compared to the same difference in the pre-crisis period (column 1).[9] Using the average decline in GDP growth of -5.5pp among the top 15 U.S. export partners between 2007 and 2009 (see Appendix Section B.1 and Table A-5), our results imply that bi-exporters exposed to a decline in foreign demand in these countries experienced about 7pp higher goods-export growth than matched mono-exporters.

[7] We use LFTTD version b201501.

[8] The tax havens are Bermuda, Cayman Islands, Hong Kong, Ireland, Luxembourg, Netherlands, Singapore, and Switzerland (Dharmapala and Hines, 2009).

[9] Appendix C.1 describes results comparing goods-export growth between matched MNE and non-MNE exporters confirming the multinational advantage indicated in aggregate statistics.

We confirm that these results are robust to an alternate control group—the same set of treated multinationals but only considering their goods exports to country $\backslash c$ where they did not sell any services in 2007. This control group accounts for unobservable firm-specific factors that determine services export status. We find that the DDD coefficient remains negative and statistically significant (column 2).

We further demonstrate in Panel B that the services export channel is driving the higher MNE goods-export growth by comparing goods-export growth to destinations with low and high services trade barriers. If services exports is the primary mechanism boosting goods-export growth, then we would expect systematically higher goods-export growth at destinations with lower barriers to services trade and the converse.

We interact the DDD coefficient (and all other terms) in Equation 1 with an indicator for whether services trade barriers are lower in country c. Using the 2008 Services Trade Restriction Indices (STRI) from the World Bank's Services Trade Restrictions Database (STRD) (World Bank, 2021a), we define barriers as low when a country's STRI score is less than or equal to the median score among all countries in the STRD and high otherwise.[10] We use both country and country-industry STRI measures differentiated by the modes of supply. The General Agreement on Trade in Services classifies trade in services supplied via four modes: cross-border provision (mode 1), consumption abroad (mode 2), commercial presence in a foreign country (mode 3), and temporary movement of people across borders (mode 4) (Francois and Hoekman, 2010). Mode 1 is the most prevalent mode of supplying the services collected in the BE-125 survey.

The quadruple interaction term compares goods-export growth at bi- and mono-exporters in response to changes in GDP growth in countries with low services trade barriers than in countries with high services trade barriers, before and after the financial crisis. This coefficient is negative across all three columns and statistically significant for restrictiveness indicators based on mode 1 across all industries and all modes in the professional services industry. These results confirm that the bi-exporter goods-export growth advantage varies systematically with the services trade restrictiveness regimes in the export markets.

We examine the timing of impacts by replacing $I(> 2008Q3)$ with quarter-specific interactions in Equation (1). The first column in Table 3 shows that the DDD coefficient is statistically significant beginning in the second quarter of 2009. Additionally, we check the robustness of the findings to using quarterly variation in GDP growth. This information is

[10]The STRI ranges between 0 and 100 where a higher value indicates higher restrictiveness of policies and regulations that govern foreign services or foreign service providers in a country (Borchert, Gootiiz and Mattoo, 2012). See Appendix Section B.2 and Figure A-1 for additional details.

available for a smaller subset of 45 countries from the OECD (Organisation for Economic Co-operation and Development, 2021). The second column shows that the timing of the impact is robust to using quarterly GDP.

5 Mechanisms

We examine whether one-way demand complementarity between goods and services is the main driver of the baseline result. We first document a bi-exporter goods quality premium, proxied by higher unit value (see Appendix Section C.2 and Table A-7), as supportive evidence of services exports acting as a demand shifter. Ideally, we would use a direct mapping between different services and the good that is its essential complement to test whether the baseline results are stronger for the identified goods. In the absence of such a mapping, we proceed in two steps. First, we identify the types of services most strongly associated with goods-export growth. Next, guided by these results, we implement a novel measure classifying goods that are most likely to exhibit one-way complementarity with the type of services driving the baseline result.

Type of Services and Goods-Export Growth

To understand which types of services are most strongly associated with goods-export growth, we replace $MNE^{SVC}_{ic,2007}$ in Equation 1 with the type of service exported by firm i to country c as a share of the firm's total services exports in 2007. In column (1) of Table 4, Panel A, we find that the DDD coefficient is largest and statistically significant at the 1% level for exports of services related to industrial IP that includes rights related to patents, trade secrets, trademarks, computer software, franchise fees, and other intellectual property and intangibles. In the next four columns, we decompose services exporters by services sales to unaffiliated or affiliated parties abroad. Columns (2) and (4) show that bi-exporters that sold services to unaffiliated and affiliated buyers, respectively, experienced higher goods-export growth; however, the result is statistically significant for services exports to affiliated parties.[11] Columns (3) and (4) reinforce the results in column (1): exports of industrial IP is most strongly associated with higher bi-exporter goods-export growth.[12]

[11] Bi-exporters in columns (2) and (4) are not mutually exclusive because they may sell services through both modes.

[12] Affiliated goods-export growth also exhibits a positive and statistically significant relationship with Media IP but the association is about half that with Industrial IP. For a 1pp decline in GDP growth, evaluated at the mean (Table 1), the effect of Media IP is (-35.33 x 0.01 x 1)=-0.35 and Industrial IP is (-2.13 x 0.35 x 1)=-0.75. Media IP may include proprietary management materials shared between subsidiaries.

We do not observe the individual goods or services buyer. However, our focus on MNEs' services exports to the *same* foreign market as their goods exports suggest that bi-exporters may be selling services to their goods buyers directly as well as through their foreign affiliates located in the export destination.

High-Intangible Products and Industrial IP

Guided by the strong role of industrial IP augmenting bi-exporters' goods-export growth during the financial crisis, we classify a set of goods most likely to exhibit one-way essential complementarity with services like industrial patents, trademarks, computer software. We rely on a product classification system that indicates the technology or software embedded in a good—the Export Control Classification Number (ECCN)—used in the U.S. export control system (Bureau of Industry and Security, 2025).

An ECCN, a five-digit alphanumeric code, categorizes items based on the *nature* of the product i.e. type of commodity, software, or technology. For example, software classified by ECCN 4D993 include Cisco System Inc.'s "EN911-NETSER" that configures VPNs (Cisco Systems Inc., 2023b); ECCN 5D992.c include Apple Inc.'s "FaceTime" (Apple Inc., 2023); Microsoft Corporation's "Office 365"; Oracle Corp.'s "Oracle Database" (Oracle Corp., 2023). A controlled product's technical parameters and intangible characteristics like software or technology are controlled based on the hardware. We define high-intangible goods as 10-digit Schedule B goods associated with a software or technology ECCN. Appendix Section B.3 and Table A-8 provide details on ECCNs and our classification methodology.

In Table 4, we compare bi- and mono-exporters' growth in high-intangible (column 6) and in non-high intangible (column 8) goods exports. The DDD coefficient is negative and statistically significant for high-intangible goods and orders of magnitude larger than for non-high intangible goods. The DDD coefficients by the type of service are reported in columns (7) and (9), and confirm that industrial IP is associated with the large, positive, and statistically significant association with growth in exports of high-intangible goods. Tests for the equality of the DDD coefficients in columns (6) and (8) rejects the null hypothesis at the 10% level; and (ii) DDD coefficients associated with Industrial IP in columns (7) and (9) rejects the null hypothesis at the 5% level. These results provide strong evidence for one-way complementarity between services and goods driving bi-exporters' differential goods-export growth during the global financial crisis.

Testing the Joint Production Hypothesis

Industrial IP export status may in part reflect the joint nature of bi-exporters' production technology where industrial IP are inputs in goods production. For example, Ding (2023) shows that "a positive demand shock in one industry of a firm increases its sales in another the more that the two industries share knowledge inputs". These knowledge inputs include rights related to patents, trademarks, trade secrets, computer software. We may thus observe a positive association between exports of industrial IP even in the absence of one-way demand complementarities.[13]

We construct a measure of proprietary knowledge produced within the firm boundary as the number of workers at a firm's auxiliary establishments. Ding, Fort, Redding and Schott (2022) show that auxiliaries—service establishments that primarily serve other establishments of the firm including R&D labs and warehouses—are generators and repositories of firm-specific intangible capital. We identify bi- and mono-exporters in our baseline sample that have at least one auxiliary establishment and re-estimate Equation 1. Hence, we compare firms that share some degree of jointness in production between knowledge inputs and goods. If bi-exporters' joint production technology is the main driver of our baseline finding, then we would not expect to find a statistically significant relationship between bi-exporter status and goods-export growth at firms that jointly produce goods and in-house proprietary knowledge.

We present the results in the last two columns of Table 4. Column (10) reports the overall DDD coefficient and column (11) reports the DDD coefficient separated by types of services exports. Our baseline findings persist strongly in this sample of firms confirming the role of Industrial IP exports above and beyond the joint nature of producing high-intangible goods and knowledge inputs.

6 Evidence of Servitization at U.S. Multinational Firms

We explore two implications of our main finding—bi-exporters' higher goods-export growth stems from exports of industrial IP to both unaffiliated and affiliated firms—for servitization of U.S. multinationals: bi-exporters' foreign affiliate services sales growth and bi-exporters' U.S. employment growth by sector.

[13]Demand complementarities may also be driven by the joint nature of the foreign buyer's production technology where they demand both industrial IP and high-intangible goods to produce a final output. Since we do not observe the foreign goods buyer in an export transaction, we cannot evaluate the nature of the foreign buyer's production technology as a source of demand complementarity.

Increase in Bi-exporters' Services Sales Growth at Foreign Affiliates

The evidence thus far suggests that higher bi-exporter goods-export growth is associated with exporting services directly to the goods buyers as well as through foreign affiliates' sales in the buyers' markets. We do not observe foreign affiliates' individual buyers, however, we can distinguish foreign affiliates' services sales to local or other non-U.S. markets.[14] We estimate,

$$\begin{aligned}\Delta_{DHS}\text{Services Sales}_{icy} = \beta_0 &+ \beta_1 \Delta \ln GDP_{cy} \times MNE^{SVC}_{ic,2007} \times Post \\ &+ \beta_2 \Delta \ln GDP_{cy} \times MNE^{SVC}_{ic,2007} + \beta_3 MNE^{SVC}_{ic,2007} \times Post \\ &+ \beta_4 MNE^{SVC}_{ic,2007} + \alpha_{iy} + \alpha_{cy} + \varepsilon_{icy}, \end{aligned} \tag{2}$$

where Δ_{DHS} is annual growth rate measured following Davis, Haltiwanger and Schuh (1996). The DHS growth rate enables a decomposition of growth in foreign affiliate's sales by market such that the coefficient β_1 in Equation 2 can be decomposed into $\sum_{j \in J} \beta_1^j$.[15] *Post* is an indicator that takes on a value of 1 after 2008 or 0 otherwise.

The first three columns in Table 5, Panel A, focus on sales to unaffiliated parties. If MNEs rely on their foreign affiliates to facilitate services sales for their goods buyers, then we would expect to find an impact on foreign affiliates' sales to unaffiliated parties relative to sales to the parent or other foreign subsidiaries. Across all three columns, the DDD coefficient is negative and statistically significant. Column (1) shows that a 1pp decline in GDP growth is associated with a 5.3pp higher growth in bi-exporters' foreign affiliate services sales. The decomposition in columns (2) and (3) indicates that this is almost entirely driven by growth in local sales.

The last three columns focus on total services sales by combining affiliated and unaffiliated services sales. We find that growth in total services sales by foreign affiliates was higher at bi- than mono-exporters during the financial crisis (column 4). The impact on affiliated sales is small, (-5.797 - (-5.264))=0.533, providing further support that the growth in foreign affiliates' services sales was driven primarily by sales to unaffiliated parties and not to other subsidiaries. The decomposition in columns (5) and (6) corroborate that the growth in local sales drove total services sales growth.

[14] We use the 2006-2008 Annual Survey of U.S. Direct Investment Abroad (Form BE-11) and 2009 Benchmark Survey of U.S. Direct Investment Abroad (Form BE-10) to obtain annual services sales by U.S. parents' foreign affiliates. U.S. parents report foreign affiliates' sales to affiliated and unaffiliated parties in: (i) affiliates' country (i.e., local), (ii) U.S., and (iii) all other non-U.S. countries.

[15] $\Delta_{DHS} X_t^j \equiv \frac{X_t^j - X_{t-1}^j}{0.5(X_t + X_{t-1})}$ where $\sum_{j \in J} X_t^j = X_t$ and j is a market. Although we do not report results for sales to the U.S. market, it can be inferred by subtracting the coefficient on the total measure from the coefficients on the local and other measures.

Increase in Bi-exporters' U.S. Services Employment Growth

We examine quarterly domestic employment growth at bi- and mono-exporting U.S. parents that sold high-intangible goods by estimating,

$$\begin{aligned}\Delta_{DHS} Emp_{iq} = \beta_0 &+ \beta_1 Exposure_{iy} \times MNE^{SVC}_{i,2007} \times I(> 2008Q3) \\ &+ \beta_2 Exposure_{iy} \times MNE^{SVC}_{i,2007} + \beta_3 MNE^{SVC}_{i,2007} \times I(> 2008Q3) \\ &+ \beta_4 Exposure_{iy} \times I(> 2008Q3) + \beta_5 Exposure_{iy} + \alpha_i + \alpha_q + \varepsilon_{iq}. \qquad (3)\end{aligned}$$

Quarterly employment for 2006-2009 is obtained from the County Business Patterns Business Register (CBPBR) which contains records of all U.S. establishments with positive payroll and used to construct the LBD (Chow et al., 2021). $Exposure_{iy}$ is a firm's exposure to the financial crisis and constructed as follows,

$$Exposure_{iy} = \sum_{c \in C_i} \left\{ (\Delta \ln GDP_{cy} - \Delta \ln GDP_{c,y-1}) \frac{Exports_{ic}}{Exports_i} \right\},$$

where C_i is the set of countries to which firm i exported goods in 2007. Thus, for each year in our sample we take change in country c's annual GDP growth and weight it by firm i's share of goods exports to c averaged over the pre-crisis period 2005–2007. A one unit change in ($Exposure_{iy}$ x 100) is the percentage-point change in annual GDP growth weighted by firm-specific export shares.[16] The average MNE in the sample experienced a -1.9pp GDP growth across their pre-crisis export markets (Table 1).

The DDD coefficient, β_1, in Equation 3 is reported in Table 5, Panel B. Results in the first column indicate that overall employment growth was not statistically significantly different between bi- and mono-exporters in the quarters during the crisis compared to the same difference in the pre-crisis quarters. We decompose firm employment growth into growth in manufacturing, services, and all other sectors, respectively in the next three columns.[17] We find that bi-exporters exhibit higher employment growth in the services sector compared

[16] $Exposure_{iy}$ captures a country's GDP growth trajectory. Suppose Country A experienced a larger decline in GDP growth rate from 10% to 5% than Country B which experienced a smaller decline in GDP growth of 2% to 1%, before and after the financial crisis. Our exposure measure will capture that a firm who exports to Country A is more exposed to the financial crisis than a firm who exports to Country B because of the larger adverse effect of the financial crisis on Country A's GDP growth. Using GDP growth rates would instead have grouped firms exporting to Country B as being more exposed compared to Country A.

[17] Industries are defined using two-digit the North American Industry Classification System (NAICS): manufacturing (31–33); services (51, 52, 53, 54, 55, 56, 61, 62, 71, 72, 81); and other (11, 21, 22, 23, 42, 44–45, 48–49) (U.S. Census Bureau, 2025).

to mono-exporters; and there is no statistical difference between bi- and mono-exporters' employment growth in manufacturing or other sectors. We further decompose growth in services employment, in the last two columns, into growth in professional and technical services (NAICS 54) and information (NAICS 51), which correspond most closely to the services collected on the BE-125 (Bruner and Grimm, 2019), and other services sectors. The positive services employment growth differential between bi- and mono-exporters is driven by growth in employment in NAICS 54 and 51.

7 Conclusion

We compare the performance of multinationals that export both goods and services to multinationals that only export goods during the 2008-2009 financial crisis using newly available links between the universe of U.S. goods traders and comprehensive surveys on U.S. multinationals and services traders. Employing a triple-difference identification strategy combined with propensity-score matching, we find that bi-exporters experienced higher goods-export growth compared to mono-exporters in response to a decline in foreign demand during the crisis. This advantage is primarily driven by exports of industrial intellectual property rights, such as patents and trademarks. We find evidence of one-way demand complementarity as a key mechanism underlying the MNE goods-export growth advantage. We also document higher growth in bi-exporters' foreign affiliate services sales and domestic employment in services sectors.

These results suggest that services exports played a crucial role in supporting foreign demand for U.S. goods during the crisis and underscores an opportunity for export promotion agencies (EPA) to target new avenues for international market expansion. EPA programs can be especially effective when exporters face high trade barriers abroad (Lederman, Olarreaga and Payton, 2010), but, have traditionally focused on promoting goods exports. While services trade between high-income countries do not face many restrictions, services trade barriers can be especially high in fast-growing Asian countries like China, India, Indonesia, Malaysia, Philippines, and Thailand, as well as in the Middle East (Borchert, Gootiiz and Mattoo, 2014). Our results suggest that expanding the scope of EPAs to integrate programs promoting services trade would potentially expand the markets for both goods and services exports.

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Table 1. Summary Statistics of U.S. Multinational Exporters

Variable	Mean	SD
Service type as a share of total services exports:		
Industrial-related Intellectual Property	0.347	0.435
Media-related Intellectual Property	0.010	0.088
Business Services	0.514	0.443
Telecommunication	0.095	0.253
Other	0.039	0.172
Goods-export growth	-0.015	1.376
Financial crisis exposure	-0.019	0.017
U.S. parent's domestic employment growth in:		
All industries	-0.031	0.300
Manufacturing	-0.028	0.204
Services	-0.004	0.154
Other	0.001	0.172
U.S. parents' foreign affiliate services sales growth to:		
All parties	0.102	0.973
All unaffiliated parties	0.069	0.849
Local unaffiliated parties	0.076	0.780
Other unaffiliated parties	0.002	0.190

Notes: This table displays the mean and standard deviation (SD) of select firm-level variables.
Source: Authors' calculations using 2007 BE-11, BE-12, BE-125, 2006-2009 LFTTD, CBPBR.

Table 2. The Role of Services Exports in U.S. Multinational Goods-Export Growth

Panel A: Baseline DDD Regressions			
Control MNE:	Goods-Only	Treated in $\setminus c$	
$\Delta GDP \times MNE^{SVC}_{ic,2007} \times I(> 2008Q3)$	-1.180**	-1.197*	
	(0.466)	(0.651)	
Fixed Effects	*firm-quarter, country-quarter, firm-region*		
Observations[a]	158,000	137,000	
Panel B: Identification Checks			
Control MNE:	Goods-Only		
$\Delta GDP \times MNE^{SVC}_{ic,2007} \times I(> 2008Q3)$	-0.535	-0.864	-0.453
	(0.665)	(0.896)	(0.635)
×Services Trade Restrictiveness Indicator			
Mode 1, All Industries	-3.297**		
	(1.389)		
Mode 1, Professional Services		-0.265	
		(1.197)	
All Modes, Professional Services			-3.145**
			(1.300)
Fixed Effects	*firm-quarter, country-quarter, firm-region*		
Observations[a]	140,000	140,000	140,000

Notes: * p<10%; ** p<5%; *** p<1%; robust standard errors in parentheses clustered at the country level. Panel A displays results from estimating Equation (1). Panel B displays results from estimating Equation (1) interacted with indicators of whether a country has low barriers to services trade. Services trade restrictiveness indicators described in Section 4 and Appendix Section B.2. The dependent variable is a firm's arm's-length goods-export growth at the country-quarter level.

[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.

Source: Authors' calculations using 2007 BE-11, BE-12, BE-125, 2006-2009 LFTTD.

Table 3. The Role of Services Exports in U.S. Multinational Goods-Export Growth: Timing of Events
Control Group: Multinational Goods-Only Exporters

	Annual GDP	Quarterly GDP
$\Delta GDP \times MNE^{SVC}_{ic,2007}$ ×Quarterly Indicator		
×2008Q4	0.373	-0.909
	(1.266)	(1.268)
×2009Q1	-0.972	-0.870
	(0.675)	(0.621)
×2009Q2	-1.751**	-1.533*
	(0.777)	(0.887)
×2009Q3	-1.362*	-1.305*
	(0.764)	(0.732)
×2009Q4	-1.429**	-1.656**
	(0.720)	(0.737)
Fixed Effects	*firm-quarter, country-quarter, firm-region*	
Observations[a]	131,000	111,000

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered at the country level. This table displays the timing of events by quarter with 12-month GDP growth using annual (column 1) and quarterly (column 2) measures. The dependent variable is a firm's arm's length goods-export growth at the country-quarter level.

[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.

Source: Authors' calculations using 2007 BE-11, BE-12, BE-125, 2006-2009 LFTTD.

Table 4. The Role of Services Exports in U.S. Multinational Goods-Export Growth: Evidence of One-way Demand Complementary

Control Group: Multinational Goods-Only Exporters

	All	Unaffiliated		Affiliated		High Intangible Goods		non-High Intangible Goods		Has Auxiliary Establishments	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
$\Delta GDP \times MNE^{SVC}_{ic,2007} \times I(>2008Q3)$		-0.995		-1.065**		-2.295**		-0.249		-1.057*	
		(1.047)		(0.525)		(1.145)		(0.618)		(0.618)	
$\Delta GDP \times I(>2008Q3) \times \frac{Service\ Type_{ic,2007}}{Services\ Exports_{i,2007}}$											
Industrial IP	-2.898***	-4.135***		-2.131***		-4.134***		-0.811		-2.919***	
	(0.692)	(1.449)		(0.751)		(1.056)		(0.966)		(0.808)	
Media IP	-1.020	-0.115		-35.33**		-		-		-	
	(2.581)	(2.682)		(14.46)		-		-		-	
Business	-0.054	-0.088		-0.417		-2.093		-1.069		0.934	
	(1.128)	(1.199)		(2.032)		(1.572)		(1.708)		(1.658)	
Telecommunication	0.761	1.417		1.587		-1.370		1.507		-1.525	
	(2.246)	(3.232)		(1.680)		(3.202)		(4.421)		(2.169)	
Other	-0.249	0.611		-0.272		-0.655		2.650		0.054	
	(1.806)	(1.866)		(2.890)		(2.467)		(3.372)		(1.945)	
Fixed Effects	*firm-quarter, country-quarter, firm-region*										
Observations[a]	158,000	54,500	54,500	134,000	134,000	63,500	63,500	130,000	130,000	106,000	106,000

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered at the country level. This table displays results from estimating Equation (1) on select sub-samples. The dependent variable is a firm's arm's-length goods-export growth at the country-quarter level in all goods by all firms (column 1); all firms with services exports to unaffiliated (columns 2, 3) and affiliated (columns 4, 5) parties; in high-intangible goods by all firms (columns 6, 7); in non-high intangible goods by all firms (columns 8, 9); and all goods by firms with at least one auxiliary establishment (columns 10, 11). Service types defined in Section 3; *Other* includes *Media IP* in columns 7, 9, 11. High-intangible goods defined in Section 5.

[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.

Source: Authors' calculations using 2007 BE-11, BE-12, BE-125, 2006-2009 LFTTD.

Table 5. Services Activities of U.S. Parents
Control Group: Multinational Goods-Only Exporters

	Unaffiliated Sales			Total Sales		
Panel A: Foreign Affiliate Services Sales Growth	All Markets	Local Market	Other	All Markets	Local Market	Other
$\Delta GDP \times MNE^{SVC}_{ic,2007} \times Post$	-5.264**	-5.844***	0.277	-5.797***	-6.548***	0.693
	(2.009)	(2.112)	(0.411)	(2.137)	(2.347)	(0.765)
Fixed Effects	*firm-year, country-year*					
Observations[a]	2,800	2,800	2,800	2,800	2,800	2,800
Panel B: Domestic Employment Growth						
					Services	
	Total	Manufacturing	Services	Other	NAICS 51 & 54	All Others
$\Delta Exposure \times MNE^{SVC}_{i,2007} \times I(> 2008Q3)$						
	-0.663	-0.035	-0.963**	0.335	-0.781**	-0.182
	(0.768)	(0.521)	(0.467)	(0.444)	(0.362)	(0.319)
Fixed Effects	*firm, quarter*					
Observations[a]	12,000	12,000	12,000	12,000	12,000	12,000

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered at the country level. Panel A displays results from estimating Equation (2). The dependent variable is a firm's annual foreign affiliate services sales growth at the country level. Panel B displays results from estimating Equation (3). The dependent variable is a firm's employment growth at the quarter level. NAICS 51 "Information" and NAICS 54 "Professional, Scientific, and Technical Services".

[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.

Source: Authors' calculations using 2006, 2008, and 2009 BE-11, 2007 BE-10, LFTTD in Panel A; and 2007 BE-11, BE-12, BE-125, 2006-2009 CBPBR, 2005-2007 LFTTD in Panel B.

A Appendix: Propensity Score Matching

Propensity Score Model

We estimate the probability that a multinational goods exporter also exports services before the global financial crisis, in 2007, using the following probit model:

$$MNE_i^{SVC} = \Phi^{-1}\Bigg[\beta_1 \ln\left(\frac{Sales_i}{Employment_i}\right) + \beta_2 \ln\left(\frac{Payroll_i}{Employment_i}\right) + \beta_3 Age_i \tag{4}$$
$$+ \beta_3 \ln Products_i + \beta_4 \ln Countries_i + \beta_5 \ln Exports_i + \beta_6 \left(\frac{ATPExports_i}{Exports_i}\right) + \varepsilon_i\Bigg],$$

where Φ is the cumulative distribution function of the standard normal distribution. MNE_i^{SVC} indicates whether a firm i is a bi-exporter. We select explanatory firm-level variables guided by prior work and measured as of 2007 as follows:[18]

- $\ln\left(\frac{Sales_i}{Employment_i}\right)$ is labor productivity;
- $\ln\left(\frac{Payroll_i}{Employment_i}\right)$ is average pay per worker;
- Age_i is the number of years the firm has been in business;
- $\ln Products_i$ is the log total number of 4-digit Schedule B exported products;
- $\ln Countries_i$ is the log total number of export destinations;[19]
- $\ln Exports_i$ is the log of total goods exports;[20]
- $\left(\frac{ATPExports_i}{Exports_i}\right)$ is the value of exports of advanced technology products (ATP) in the share of the firm's total exports to capture its underlying production technology (U.S. Census Bureau, 2021a).

Firm payroll and employment is obtained from the 2007 Longitudinal Business Database (LBD) which tracks U.S. establishments in the non-farm, private sector employer universe over time (Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson and White, 2021).[21] We classify a firm's primary industry, a six-digit NAICS, as the industry that accounts for the largest share of its employment. Then we define a two-digit sector based on the firm's

[18] Alfaro and Chen (2012) include an establishment's sales, labor productivity, age, multi-product status, exporter status; Alviarez et al. (2017) include establishment sales, age, and multi-product status.

[19] Tax havens are excluded from the analysis.

[20] Multinationals' total exports include arm's length and related-party trade. Related-party relationships are those in which one firm owns a stake of at least 10% in the other (U.S. Census Bureau, 2020).

[21] LBD excludes operations with no statutory employees, e.g. self-employed, farms (but not agri-business), and the public sector. It contains information on every establishment's firm affiliation, year of birth, six-digit NAICS, employment, and payroll.

primary NAICS. Information on firm exports is obtained from the 2007 LFTTD. We obtain total firm sales from the 2007 Economic Census. The Economic Census, conducted every 5 years, is the most comprehensive survey of businesses operating in the United States (U.S. Census Bureau, 2022b).[22]

Matching MNE Goods-Only Exporters

We begin with the sample of all multinationals that export goods in 2007. Then we restrict to only include mono-exporters that export (i) at least one 4-digit Schedule B good as bi-exporters; and (ii) to at least one destination as bi-exporters.[23] After applying these restrictions, there are 6,700 firms that account for US$383 billion of goods exports.

We further restrict to goods exporters that have (i) arm's length exports in 2006 and 2007 to calculate pre-crisis export growth; and (ii) information on employment and sales. After applying these restrictions, there are 5,200 firms that account for $359 billion of goods exports.

Then we estimate Equation (4) and obtain propensity scores for all multinational firms in this sample. Equation 4 is estimated separately for each 19 two-digit NAICS sectors and further separated by parent and affiliate status of MNEs to obtain the set of MNE goods exporter control firms. This ensures that we match U.S. parent bi-exporters with U.S. parent mono-exporters only and foreign parent bi-exporters with other foreign parent mono-exporters only. We implement nearest neighbor matching without replacement. The matched sample contains 750 bi-exporter and 750 mono-exporters, together, accounting for $180 billion of goods exports.

We assess how well the propensity score matching performs by the resulting balancing quality between the treated and control groups. First, we check the difference in the means of the explanatory variables in the propensity score model between the treated and control firms in Panel A of Table A-1. We find no statistically significant differences between the treated and control firms in terms of pre-crisis characteristics of labor productivity, number of exported products, or number of export destinations. The difference in means of the size measure (total value of goods exports) is only marginally significant with a p-value of 0.09. Second, the Rubin's B, the standardized difference in the linear index of the propensity score, is 8.7 and the Rubin's R, the ratio of the propensity score's variance in the treated and untreated sample, is 1.45. Finally, the average difference in the propensity scores of the matched firms in our sample is 0.023. Together, these results lend confidence that our matching procedure has grouped together firms with similar observable characteristics.

[22]The Economic Census includes: Census of Manufactures; Census of Services; Census of Construction; Census of Transportation, Communications, and Utilities; Census of Mining; Census of Retail Trade; Census of Wholesale Trade; Census of Finance, Insurance, and Real Estate (U.S. Census Bureau, 2022a).

[23]We do not restrict a mono-exporter to sell the same 4-digit Schedule B good to the same destination as the matched bi-exporter due to sample attrition.

Characteristics of Matched MNEs

Using the universe of multinationals (identified using BE-11 for domestic MNEs and BE-12 for foreign MNEs) in 2007 linked to LFTTD, multinational firms accounted for 67% of total goods exports. Notably, MNEs that also exported services accounted for 47% of overall goods exports; while MNEs that only exported goods accounted for only 20% of overall exports.

Table A-2 shows the share of employment by broad sectors of bi-exporters (column 1) and mono-exporters (column 2) in 2007.[24] We document that about a third of employment is in the wholesale, retail, and transportation sectors for both bi- and mono-exporters. Bi-exporters have a slightly smaller share of employment in manufacturing (25% versus 28%) while they tend to have higher shares of employment in information (7% versus 4%), and professional services (8% versus 5%). Mono-exporters have higher shares of employment in finance, insurance, real estate (12% versus 6%).

We also document that bi-exporters have a higher share of its manufacturing employment in sectors that produce advanced technology products. We identify four-digit manufacturing NAICS that map to the ten different ATP classifications (Goldschlag and Miranda, 2020). As shown in the last row of Table A-3, 58% of bi-exporters' employment in the manufacturing sector is in ATP and this share for mono-exporters is only 37%. Looking within ATP categories, we find bi-exporters have higher shares of employment in aerospace (10% versus 5%), information and communications (11% versus 6%), life sciences (10% versus 7%), and optoelectronics (8% versus 4%).

B Appendix: Data and Measurement

B.1 Source of Variation in GDP Growth

Table A-5 displays the annual GDP growth rate for the top 15 export partners in three time periods: 2006–2007, 2007–2008, and 2008–2009.[25] These countries accounted for 70% of total U.S. exports in 2007. The first columns shows that the annual growth rate between 2006 and 2007 is positive for all partner countries. In the second column, the growth rate between 2008 and 2007 is positive for all countries except for Italy, Japan, and the United Kingdom. Between 2008 and 2009, shown in the last column, 10 out of the 15 trading partners experienced negative GDP growth, with the exception of Australia, China, India, Israel, and South Korea. However, with the exception of India, even the countries with a positive GDP growth rate experienced a decline in their GDP growth between 2007 and 2009 as seen across the three columns. We use the variation in annual GDP growth across all U.S. partner countries to measure the differential exposure of U.S. exporters to foreign demand.

[24] See Jensen (2011) for industrial activities of all U.S. services importers and exporters.

[25] The base period in the GDP growth calculation is $t-1$.

B.2 Services Trade Restrictiveness Index

We utilize the STRI from the World Bank's Services Trade Restrictiveness Database (STRD) to create indicator variables that categorize a country as having low or high barriers to services trade as: (i) mode 1 across all industries; (ii) mode 1 in the professional services industry; and (iii) all modes in the professional services industry. We use 2008 STRI since it is the first available year in the STRD. We use these measures for two reasons. The BE-125 survey collects information on U.S. persons' direct transactions, both sales and purchases, with affiliated and unaffiliated foreign person performed "on a cross-border basis" (mode 1) and "in person" (mode 4).[26] Aggregate estimates for the U.S. show that aside from mode 3 (which is largely excluded in the BE-125 survey), mode 1 is the most prevalent mode of supplying services (see Figure 2 in Mann (2019)); for specific types of services collected in the BE-125, such as charges for intellectual property and business and management consulting categories, cross-border trade is the primary mode of supplying the service (see Figure 3 in Mann (2019)). We separately consider STRI for the professional services (legal, accounting, engineering, and architecture) industry because this industry corresponds most closely with activities associated with the types of services collected on the BE-125 survey instrument.[27]

Our indicator for a low STRI score is equal to 1 when a country's score is less than or equal to the median score among all countries in the STRD. A low STRI score indicates that there are fewer restrictions faced by U.S. firms exporting services to these countries. We present the underlying STRI data used to construct these three separate indicators in Appendix Figure A-1. Considering different industries or modes of supply change whether a country is considered to have low (below the median line) or high (above the median line) restrictiveness. For example, focusing on mode 1 supply but comparing the STRI score for all industries (Figure A-1(a)) to professional services (Figure A-1(b)), we find that countries switch their restrictiveness rankings: Mexico moves from high to low while United Kingdom and Germany move from low to high.

B.3 Classifying High-Intangible Products

This section describes our methodology to classify high-intangible goods—the set of 10-digit Schedule B products that are most likely to exhibit one-way essential complementarity with industrial IP. One-way complementarity implies that patents, trademarks, computer software entail the use of a related good to raise the utility of the foreign buyer.

We identify high-intangible goods using a unique feature of the U.S. export control system that restricts exportation of dual-use items to certain destinations (Congressional Research Service, 2021).[28] Dual-use items are commodities, software, or technologies that

[26] See definitions of reportable transactions in U.S. Bureau of Economic Analysis (2020b). Foreign affiliate sales information collected in the BE-10 and BE-11 surveys corresponds to commercial presence in countries outside the United States i.e., mode 3.

[27] We thank Jessica McCloskey at BEA for sharing the concordance between services categories in the BE-125 and NAICS.

[28] An alternate classification candidate is U.S. Census Bureau's Advanced Technology Products (ATP) that identify a select set of traded goods as containing significant amounts of advanced technologies in ten broad

have civil as well as military applications. Under the Export Administration Regulations (EAR, 15 CFR §§730-774), commercial goods subject to export controls must obtain a license prior to exportation (EAR, 15 CFR §§730-730.3).[29] An essential element in determining whether an export license is required is the classification of goods using the Export Control Classification Number (ECCN).

ECCNs are five-character alphanumeric codes only used to classify dual-use items for export control purposes.[30] An ECCN categorizes items based on the *nature* of the product, i.e. type of commodity, software, or technology and its respective technical parameters. The first character, a number, identifies a broad category—0: nuclear materials, 1: materials, chemicals, microorganisms and toxins, 2: materials processing, 3: electronics, 4: computers, 5: telecommunications and information security, 6: sensors and lasers, 7: navigation and avionics, 8: marine, 9: aerospace and propulsion; the second character identifies a product group—A: end items, B: test equipment, C: materials, D: software, E: technology.

Product groups D and E encompass software and technology, respectively. For example, 4D993 describe computer products containing "program" proof and validation "software," "software" allowing the automatic generation of "source codes," and operating system "software" that are "specially designed" for "real-time processing" equipment; and 4E993 describes computer products that contain "technology" for the "development" or "production" of equipment designed for "multi-data-stream processing." (Bureau of Industry and Security, 2023).

We start with the list of all export transactions that report ECCNs with a "D" or "E" in the 2007 LFTTD and identify the associated 10-digit Schedule B codes.[31] We then classify the exports in the Schedule B codes associated with technology and software ECCNs as high-intangible goods. Table A-8 reports the Schedule B codes that account for largest shares of exports associated with technology and software products identified using ECCNs. About a third of all exports, by value, in our sample are of software and technology Schedule B products. About 80% of value in these identified Schedule B codes are in chapters 84 ("Nuclear reactors, boilers, machinery and mechanical appliances; parts thereof"), 85 ("Electrical machinery and equipment and parts thereof; sound recorders and reproducers, television image and sound recorders and reproducers, and parts and accessories of such articles"), and 90 ("optical, photographic, cinematographic, measuring, checking, precision, medical or surgical instruments and apparatus").

Within chapter 84, 8471 ("Automatic data processing machines, magnetic or optical readers, machines for transcribing data onto data media in coded form and machines for processing such data") accounts for about a third of export value; within chapter 85, 8542 ("electronic integrated circuits") accounts for 59% of export value; and within chapter 90, 9031 ("Measuring or checking instruments, appliances and machines, profile projectors")

categories (Abbott, McGuckin, Herrick and Norfolk, 1989). ATP defines goods based on certain technologies that were used in production, *a priori*, however, this information alone is not enough to ascertain whether these goods are also essential in the use of industrial IP.

[29]License requirements depend on an item's technical characteristics, destination, end-use, and end-user.

[30]For the complete list of ECCNs see Bureau of Industry and Security (2025).

[31]There does not exist a one-to-one mapping between ECCN and Schedule B classifications: one ECCN may map into multiple Schedule B codes and vice versa as documented in Basker and Kamal (2025).

accounts for 26% of export value. We calculate the export value associated with high-intangible goods for each firm separately from export value in all other goods.

C Appendix: Additional Results

C.1 Establishing the Multinational Advantage

We examine whether the MNE goods-export growth premium in the aggregate data persists after controlling for observable firm characteristics. We compare the growth in goods exports between multinational and matched non-multinational firms. MNE exporters may have higher export growth than non-MNE exporters because they tend to be larger and more productive than non-MNEs (Antrás and Yeaple, 2014). Therefore, we construct a comparable set of non-MNE goods exporters using propensity score matching by estimating the probability that a goods exporter is a multinational.

Matching non-MNE Goods Exporters

We begin with the sample of all goods exporters in 2007. Then we restrict attention to all multinational exporters and only those non-multinational exporters whose employment is at least as large as the median multinational exporter. This restriction ensures that we are comparing firms of similar size distributions. We further restrict to non-multinationals that share (i) at least one 4-digit Schedule B good as multinationals; and (ii) to at least one destination as multinationals.[32] These restrictions minimize the possibility that differences in export growth between MNE and non-MNEs is being driven by product and country compositions. After applying these restrictions, there are 346,000 firms that account for $653 billion of goods exports.

We further restrict to goods exporters that have (i) arm's length exports in 2006 and 2007 to calculate pre-crisis export growth; and (ii) information on employment and sales. After applying these restrictions, there are 9,700 firms that account for $409 billion of goods exports.

Then we estimate Equation 4 separately for each 19 two-digit NAICS sectors to obtain the set of non-MNE goods-exporter control firms.[33] We implement nearest neighbor matching with replacement such that the same non-multinational firm may be a control for multiple multinational firms. We use five nearest neighbors to allow for larger numbers of unique controls per treatment. The matched sample contains 2,900 multinational and 2,900 non-multinational exporters together accounting for $99 billion of goods exports.

We assess how well the propensity score matching performs by the resulting balancing quality between the treated and control groups. First, we check the difference in the means of the explanatory variables in the propensity score model between the treated and control firms in Panel B of Table A-1. We find no statistically significant differences between the

[32] We do not require a non-multinational control to export the same Schedule B good to the same destination as the matched treated multinational due to sample attrition.

[33] MNE_i^{SVC} is replaced with MNE_i in Equation 4.

treated and control firms in terms of pre-crisis characteristics of labor productivity, number of exported products, or number of export destinations. The difference in means of the size measure (total value of goods exports) is statistically significant although the values are very close: 15.5 and 15.4 for MNEs and non-MNEs, respectively. Second, balancing measures based on the propensity score (Rubin, 2001) also point to high balancing quality. Rubin's B, the standardized difference in the linear index of the propensity score, is 6.9 and the Rubin's R, the ratio of the propensity score's variance in the treated and untreated sample, is close to 1 (0.96). Finally, the average difference in the propensity scores of the matched firms in our sample is 0.0002. Together, these results lend confidence that our matching procedure has grouped together non-MNE goods exporters with similar observable characteristics.

Results

We estimate the following DDD model,

$$\begin{aligned}\Delta \ln Exports_{icq} = \beta_0 &+ \beta_1 \Delta \ln GDP_{cy} \times MNE_{i,2007} \times I(> 2008Q3) \\ &+ \beta_2 \Delta \ln GDP_{cy} \times MNE_{i,2007} \\ &+ \beta_3 \overline{\Delta \ln X_{ic,2005-2007}} + \alpha_{iq} + \alpha_{cq} + \alpha_{ir} + \varepsilon_{icq}. \end{aligned} \tag{5}$$

This specification is identical to Equation 1 except that $MNE^{SVC}_{ic,2007}$ is replaced with $MNE_{i,2007}$. We examine growth in arm's-length exports and present results in Table A-6. The DDD coefficient of interest is β_1. Focusing on the first column, we find that MNE exporters facing a 1pp decline in GDP growth experienced 0.7pp higher goods-export growth than matched non-MNE exporters in the post-crisis relative to the same difference in the pre-crisis period. We decompose the timing of the MNE advantage by interacting quarter dummies with $\Delta GDP \times MNE_{i,2007}$. The second column shows that MNEs' goods-export growth is higher in all quarters beginning in 2008Q1, but it is statistically significant in 2008Q4 and 2009Q1. The third column uses quarterly GDP information for a smaller subset of countries and corroborates the findings using annual GDP.

The timing of the impact is earlier compared to results using the sample of multinationals only. MNEs are generally larger and have more resources than non-MNEs. This could explain why the MNE advantage in Table A-6 appears earlier (significant in 2008 Q4 and 2009 Q1) than in Table 3 (significant in 2009 Q2 - 2009 Q4). MNEs might have been able to respond more quickly to the initial shock due to their greater resources and global networks. Among MNEs, the specific advantage of being a bi-exporter took longer to materialize.

C.2 Bi-exporter Unit Value and Quantity

If services exports act as a demand shifter for firms by increasing the perceived quality of a firm's goods we would expect the goods to potentially exhibit higher prices as a proxy for quality (Ariu et al., 2020). We estimate the following specification to test this idea,

$$\ln y_{ipcq} = \beta_0 + \beta_1 MNE^{SVC}_{ic,2007} \times I(> 2008Q3) + \alpha_{is} + \alpha_{pq} + \alpha_{cq} + \alpha_{ir} + \varepsilon_{icq}, \tag{6}$$

where y_{ipcq} is either unit value or quantity for firm i exporting 10-digit Schedule B product p (within 2-digit Schedule B chapter s) to country c in one of seven world regions r in quarter q. β_1 captures the average difference in the unit value or quantity between bi-and mono-exporters.

Appendix Table A-7 shows that bi-exporters exhibit higher unit values and lower quantities compared to mono-exporters after the crisis relative to the same difference pre-crisis. These results are consistent with services raising the perceived quality of goods.

D Appendix: Tables and Figures

Table A-1. Balance Tests

Panel A: Matching to MNE Goods Only Exporters			
	Mean		t-tests
	$MNE^{SVC}_{i,2007}$	$MNE^{goods\text{-}only}_{i,2007}$	$p\text{-}value$
Log Sales per Worker	6.05	6.01	0.28
Log Payroll per Worker	4.21	4.17	0.12
Firm Age	27.06	26.81	0.48
Log Number of 4-digit Schedule B Products	3.50	3.42	0.15
Log Number Export Destinations	2.97	2.93	0.53
Log Exports	17.20	17.00	0.13
Advanced Technology Product Exports (%)	0.56	0.55	0.66
Firm Count	750	750	
Panel B: Matching to non-MNE Goods Exporters			
	Mean		t-tests
	$MNE_{i,2007}$	$non\text{-}MNE_{i,2007}$	$p\text{-}value$
Log Sales per Worker	5.65	5.69	0.05
Log Payroll per Worker	3.88	3.90	0.29
Firm Age	25.34	25.24	0.61
Log Number of 4-digit Schedule B Products	2.38	2.42	0.16
Log Number Export Destinations	1.94	1.96	0.64
Log Exports	14.90	14.95	0.33
Advanced Technology Product Exports (%)	0.41	0.42	0.50
Firm Count	2,900	2,900	

Notes: This table displays the means and p-values associated with t-statistics of differences in the means between treatment and control groups of variables used in the propensity score model. Panel A displays balance tests where the treatment group is multinationals that export both goods and services and the control group is multinationals that only export goods. Panel B displays balance tests where the treatment group is multinational goods exporters and the control group is non-multinational goods exporters. Advanced Technology Product defined by U.S. Census Bureau (2021a).
Source: Authors' calculations using 2007 BE-11, BE-12, BE-125, EC, LBD, LFTTD.

Table A-2. U.S. Multinational's Employment Share (%) by Broad Sectors in 2007

Broad Sectors	Multinationals that export: Goods and Services	Goods Only
Ag., Mining, Utilities, Construction	1.30	1.82
Manufacturing	25.25	27.94
Wholesale, Retail, Transportation	32.20	33.43
Information	7.39	3.90
Finance, Insurance, Real Estate	6.35	11.85
Professional, Scientific, Technical Services	8.49	5.28
Management & Administrative	12.81	12.65
Other Services	6.22	3.11

Notes: This table displays employment in eight broad sector classifications as a share of total employment at multinationals that export both goods and services (column 1) and multinationals that only export goods (column 2). "Ag., Mining, Utilities, Construction" groups NAICS 11, 21, 22, and 23; "Manufacturing" groups NAICS 31-33; "Wholesale, Retail, Transportation" groups NAICS 42, 44-45, and 48-49; "Information" is NAICS 51; "Finance, Insurance, Real Estate" groups NAICS 52 and 53; "Professional, Scientific, Technical Services" is NAICS 54; "Management & Administrative" groups NAICS 55 and 56; "Other Services" groups NAICS 61, 62, 71, 72, and 81.
Source: Authors' calculations using 2007 LBD.

Table A-3. U.S. Multinationals' Manufacturing Employment Share (%) by Advanced Technology Products in 2007

Advanced Technology Product	Multinationals that export: Goods and Services	Goods Only
Advanced Materials	6.58	4.98
Aerospace & Biotechnology	10.43	4.94
Electronics	5.14	3.35
Flexible Manufacturing	1.13	1.88
Information & Communications	11.57	6.28
Life Sciences	10.28	7.15
Nuclear	2.93	2.64
Optoelectronics	7.56	4.39
Weapons	2.67	1.75
Total ATP share in Manufacturing	58.30	37.36

Notes: This table displays employment in ten advanced technology product (ATP) categories as a share of total employment in the manufacturing sector by multinationals that export both goods and services (column 1) and multinationals that only export goods (column 2).
Source: Authors' calculations using 2007 LBD.

Table A-4. U.S. Services Exports Collected in Form BE-125, 2007

Services Product	Share of Total (%)	Share of Broad Category (%)
Industrial IP		
- Industrial Processes	7.4	44.9
- Computer Software	5.9	35.9
- Trademarks	2.4	14.3
- Franchise Fees	0.8	4.9
Total	**16.5**	**100.0**
Media IP		
- Film and Television	2.9	87.3
- Books and Sounds Recording	0.3	9.2
- Broadcasting and Recording	0.1	3.5
Total	**3.4**	**100.0**
Business		
- Management and Consulting	5.4	32.2
- R&D Testing	3.2	19.0
- Architectural and Engineering	2.3	13.9
- Operational Leasing	1.4	8.7
- Legal	1.3	8.0
- Advertising	0.8	5.0
- Construction	0.6	3.4
- Mining	0.5	3.1
- Other	0.4	2.4
- Educational and Training	0.2	1.4
- Trade-related	0.2	1.2
- Accounting	0.2	1.0
- Sports and Performing Arts	0.1	0.8
Total	**16.7**	**100.0**
Telecommunication		
- Telecommunication	1.7	40.7
- Computer and Data Processing	1.5	36.2
- Database and Other Information	0.9	23.1
Total	**4.1**	**100.0**
Other	**2.3**	**100.0**

Notes: This table displays the 2007 share of U.S. services exports collected by the U.S. Bureau of Economic Analysis in Form BE-125 "Quarterly Survey of Transactions in Selected Services and Intellectual Property with Foreign Persons". The five broad services product categories are defined in Section 3. The first column reports percentages as a share of total U.S. services exports. The second column reports percentages as a share of the total exports in a broad services product category.
Source: Authors' calculations using U.S. Bureau of Economic Analysis (2020b).

Table A-5. GDP Growth (%) of Top 15 U.S. Export Partners

Country	2006–2007	2007–2008	2008–2009
Canada	2.05	1.00	-2.97
Mexico	2.27	1.14	-5.43
China	13.31	9.21	8.98
Japan	1.64	-1.10	-5.57
United Kingdom	2.40	-0.28	-4.34
Germany	2.93	0.96	-5.86
South Korea	5.64	2.97	0.79
France	2.40	0.25	-2.92
Taiwan	6.31	0.70	-1.58
Belgium	3.61	0.45	-2.04
Brazil	5.89	4.97	-0.13
Australia	3.77	3.59	1.92
India	7.38	3.04	7.57
Italy	1.48	-0.97	-5.43
Israel	5.61	2.95	0.92

Notes: This table displays the annual GDP growth for the top 15 U.S. export partners. Countries are ranked by U.S. export value in 2007.
Source: Authors' calculations using country-level GDP from World Bank (2021b), United Nations (2021), and Federal Reserve Economic Data (2021).

Table A-6. U.S. Multinational Goods-Export Growth
Control Group: Non-Multinational Goods Exporters

	Annual GDP	Quarterly GDP
$\Delta GDP \times MNE_{i,2007} \times I(> 2008Q3)$	-0.657*	
	(0.371)	
$\Delta GDP \times MNE_{i,2007}$ ×Quarterly Indicator		
×2008$Q4$	-1.235*	-2.026***
	(0.637)	(0.710)
×2009$Q1$	-0.837**	-0.566
	(0.405)	(0.631)
×2009$Q2$	-0.415	-0.650
	(0.687)	(0.958)
×2009$Q3$	-0.641	-0.688
	(0.575)	(0.589)
×2009$Q4$	-0.405	0.181
	(0.519)	(0.626)
Fixed Effects	*firm-quarter, country-quarter, firm-region*	
Observations[a]	327,000 327,000	247,000

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered at the country level. This table displays results from estimating Equation 5. The dependent variable is a firm's arm's-length goods-export growth at the country-quarter level.
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using 2007 BE-11, BE-12, 2006-2009 LFTTD.

Table A-7. U.S. Multinational Export Premium
Control Group: Multinational Goods-Only Exporters

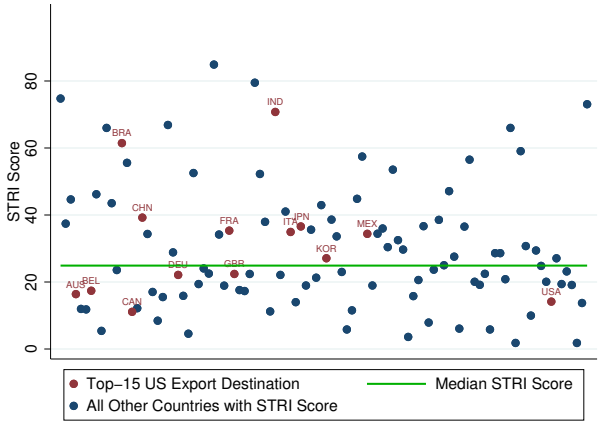
	Log Unit Value	Log Quantity
$MNE^{SVC}_{ic,2007} \times I(> 2008Q3)$	0.013*	-0.021**
	(0.008)	(0.010)
Fixed Effects	*firm-hs2, hs10-quarter, country-quarter, firm-region*	
Observations[a]	904,000	904,000

Notes: * p<10%; ** p<5%; *** p<1%. Robust standard errors in parentheses clustered at the firm level. This table displays results from estimating Equation (6).
[a] Observation counts rounded to comply with Census Bureau disclosure avoidance rules.
Source: Authors' calculations using 2007 BE-11, BE-12, BE-125, 2006-2009 LFTTD.

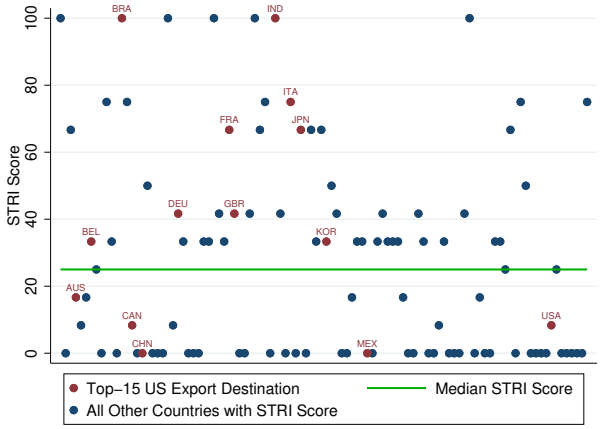
Table A-8. Classifying High-Intangible Goods, 2007

	Export Share (%)
1. Schedule B mapped to Software & Technology ECCN as a share of all exports	29.48
2. Chapter 84 as a share of (1)	28.36
Chapter 8471 as a share of (2)	32.10
3. Chapter 85 as a share of (1)	37.64
Chapter 8542 as a share of (3)	58.76
4. Chapter 90 as a share of (1)	13.46
Chapter 9031 as a share of (4)	25.81

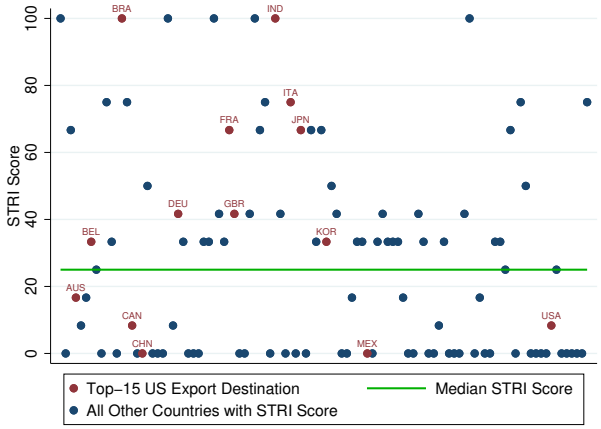
Notes: This table displays the share of exports by Schedule B products that map to Software and Technology Export Control Classification Codes (ECCN) (Bureau of Industry and Security, 2025).
Source: Authors' calculations using 2007 LFTTD.



(a) Mode 1, All Industries



(b) Mode 1, Professional Services



(c) All Modes, Professional Services

Notes: This figure plots the services trade restrictiveness indices (STRI) score based on (a) all industries using mode 1 supply; (b) professional services industry using mode 1 supply; (c) professional services industry using all modes 1 supply.
Source: Author's calculations using STRD World Bank (2021a).

Figure A-1. Services Trade Restrictiveness by Country, 2008